



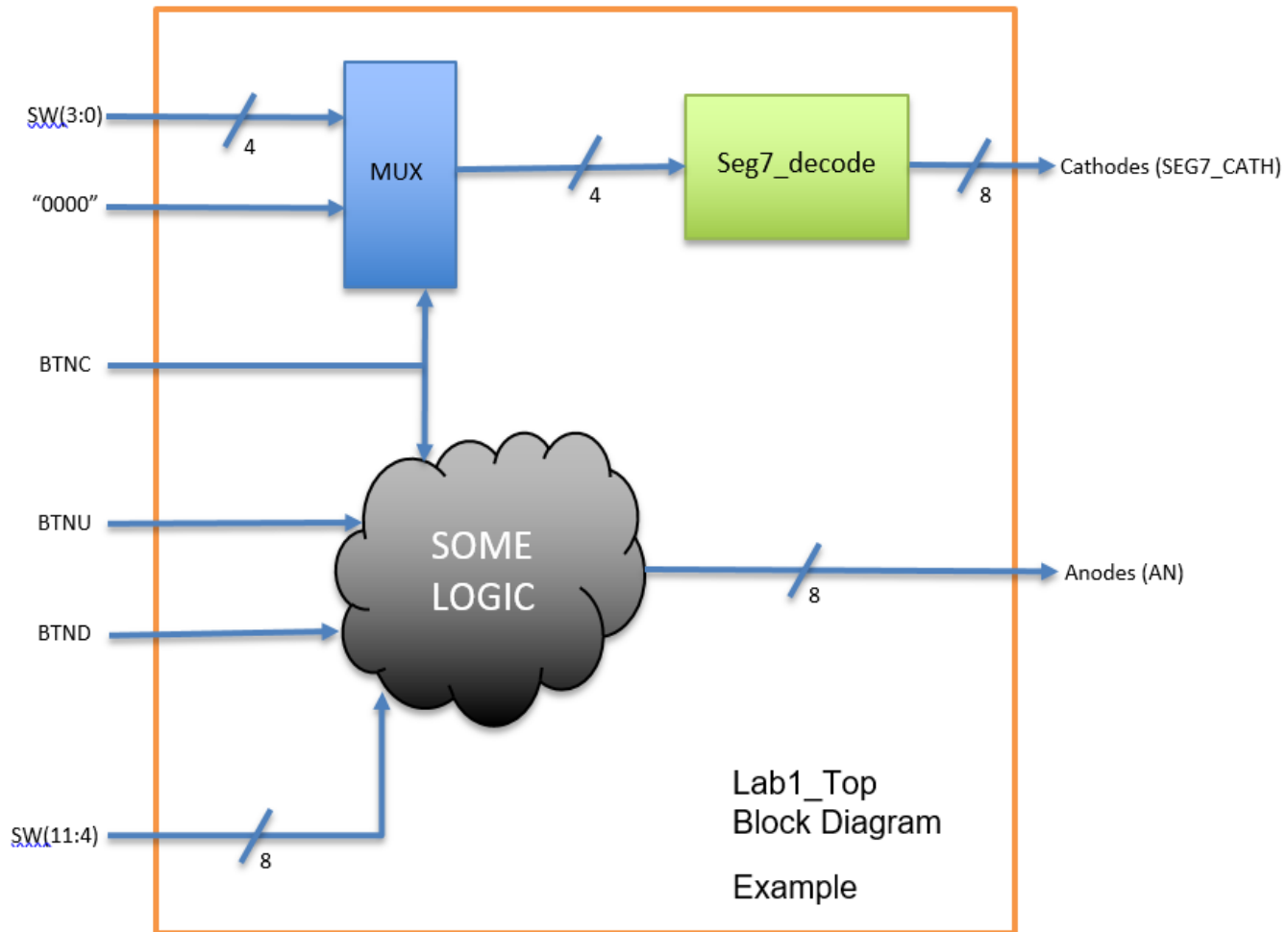
JOHNS HOPKINS

WHITING SCHOOL
of ENGINEERING

FPGA Design Using VHDL

Module 2A: Module 1 Review

Lab 1 Block Diagram



Lab 1 – Seg 7 Decode

```

entity seg7_decode is
  Port ( digit : in STD_LOGIC_VECTOR (3 downto 0);
        cathode : out STD_LOGIC_VECTOR (7 downto 0));
end seg7_decode;
architecture Behavioral of seg7_decode is
begin
  --Create the digit using the array for cathodes
  with digit select
  cathode <=
    "11000000" when x"0" ,
    "11111001" when x"1" ,
    "10100100" when x"2" ,
    "10110000" when x"3" ,
    "10011001" when x"4" ,
    "10010010" when x"5" ,
    "10000010" when x"6" ,
    "11111000" when x"7" ,
    "10000000" when x"8" ,
    "10010000" when x"9" ,
    "10001000" when x"A" ,
    "10000011" when x"B" ,
    "11000110" when x"C" ,
    "10100001" when x"D" ,
    "10000110" when x"E" ,
    "10001110" when others;

```

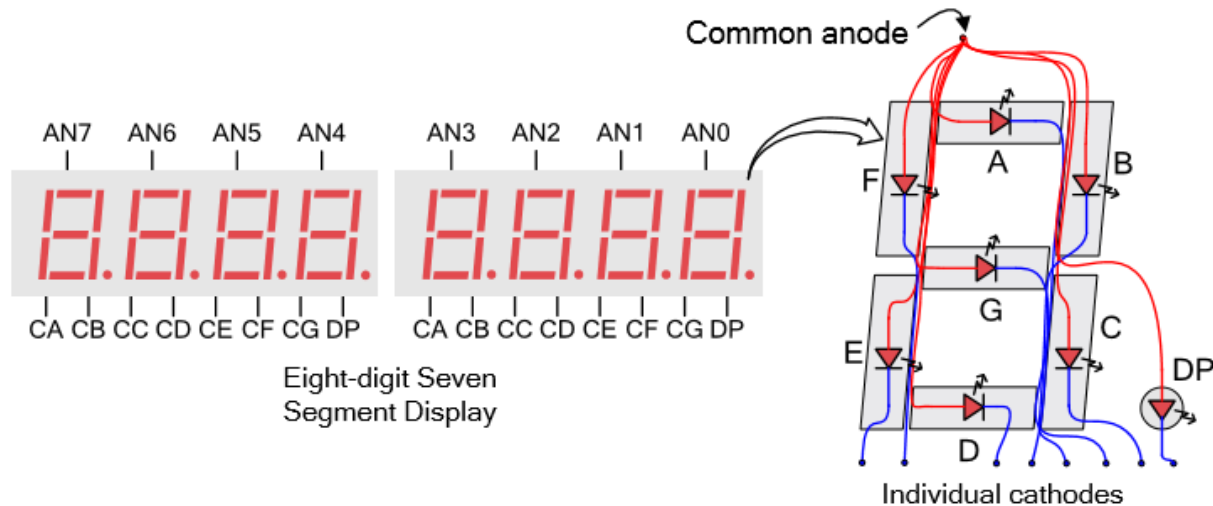


Figure pulled from Nexy's Reference Manual

```

end Behavioral;

```

Lab 1 - Top

```
entity lab1_top is
  Port (
    --Push Buttons
    BTNC : in STD_LOGIC;
    BTND : in STD_LOGIC;
    BTNU : in STD_LOGIC;
    --Switches (16 Switches)
    SW : in STD_LOGIC_VECTOR (15 downto 0);
    --LEDs (16 LEDs)
    LED : out STD_LOGIC_VECTOR (15 downto 0);
    --Seg7 Display Signals
    SEG7_CATH : out STD_LOGIC_VECTOR (7 downto 0);
    AN : out STD_LOGIC_VECTOR (7 downto 0));
end lab1_top;
```

Lab 1 Architecture

```
architecture Behavioral of lab1_top is
    signal display_digit : std_logic_vector(3 downto 0);
begin
    --The LEDs are driven directly by the value of the slider switches
    LED <= SW;

    --if the BTNC is pressed then display zero, otherwise display value on first 4 slider switches in HEX
    display_digit <= [redacted] when BTNC = '0' else [redacted]; --display 0 when center button is pressed

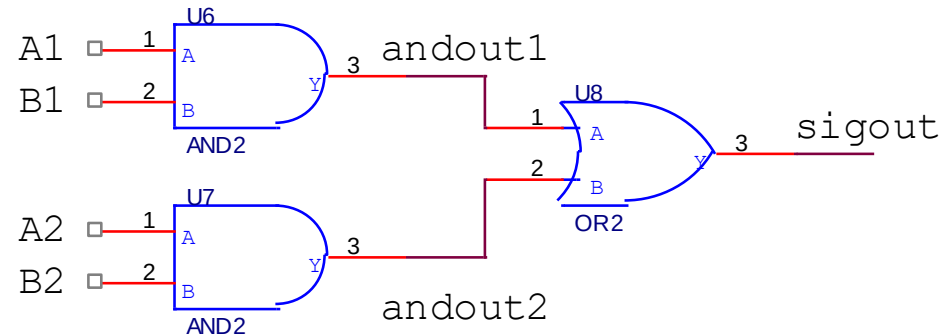
    --this decoder takes an input digit value and outputs the control signals for the cathode
    decoder : entity seg7_decode port map (digit => display_digit, cathode => SEG7_CATH);

    --Make sure BTNC is highest priority to properly engage all 0's when pressed independent of other buttons
    --Next in priority is BTNU, which when pressed will light up the upper 4 seven-seg displays
    --then when BTND is pressed it will light up the bottom 4 seven-seg displays
    --if none of them are pressed the switches will control which displays are on
    AN <= [redacted] when [redacted] else
           [redacted] when [redacted] else
           [redacted] when [redacted] else
           not [redacted];

end Behavioral;
```

Architecture Review

```
entity simplecircuit is
port (
    a1,b1,a2,b2 : in std_logic;
    sigout : out std_logic);
end simplecircuit;
```



```
architecture behavior of simplecircuit is
    signal andout1, andout2 : std_logic;
begin
```

```
-- U6 and U7 are component instantiations
sigout <= andout1 or andout2; -- This is CSA
U6: entity simpleAnd port map (a=>a1,b=>b1,y=>andout1);
U7: entity simpleAnd port map (a=>a2,b=>b2,y=>andout2);
-- All three assignments above happen concurrently (order irrelevant)
```

```
end behavior;
```

Concurrent Designs

```
begin
```

```
-- U6 and U7 are component instantiations
```

```
sigout <= andout1 or andout2; -- This is CSA
```

```
U6: entity simpleAnd port map (a=>a1,b=>b1,y=>andout1);
```

```
U7: entity simpleAnd port map (a=>a2,b=>b2,y=>andout2);
```

```
-- Process: 3rd type of concurrent design
```

```
end behavior;
```

